

Trigonometry & Polar Coordinates

ANSWER KEY



Law of Sines

- 1 In a triangle, $\angle A = 40^\circ$, $a = 10$, and $\angle B = 65^\circ$. Find b , rounded to the nearest hundredth.

By the Law of Sines, $\frac{b}{\sin B} = \frac{a}{\sin A} \Rightarrow b = \frac{10 \sin 65^\circ}{\sin 40^\circ} \approx 14.10$.

- 2 In a triangle, $\angle A = 30^\circ$, $\angle B = 75^\circ$, and $a = 6$. Find $\angle C$ first, then explain how you would find side c .

$\angle C = 180^\circ - 30^\circ - 75^\circ = 75^\circ$. Then use $\frac{c}{\sin C} = \frac{a}{\sin A}$ to solve for c .

Law of Cosines

- 3 A triangle has $a = 10$, $b = 6$, and included angle $C = 40^\circ$. Find c , rounded to the nearest hundredth.

$c^2 = a^2 + b^2 - 2ab \cos C = 100 + 36 - 120 \cos 40^\circ \approx 44.07$, so $c \approx 6.64$.

- 4 Explain when you should use the Law of Cosines instead of the Law of Sines to solve a triangle.

The Law of Cosines is used when you know two sides and the included angle (SAS), or all three sides (SSS) — cases where the Law of Sines can't be set up directly because no side-angle pair is fully known.

Double-angle identities

- 5 If $\sin x = \frac{3}{5}$ and x is in Quadrant I (so $\cos x = \frac{4}{5}$), find $\sin 2x$ and $\cos 2x$.

$\sin 2x = 2 \sin x \cos x = 2 \left(\frac{3}{5}\right) \left(\frac{4}{5}\right) = \frac{24}{25}$. $\cos 2x = \cos^2 x - \sin^2 x = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$.

- 6 If $\cos x = -\frac{1}{2}$ and x is in Quadrant II (so $\sin x = \frac{\sqrt{3}}{2}$), find $\cos 2x$.

$\cos 2x = \cos^2 x - \sin^2 x = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2}$.

Polar to rectangular conversion

- 7 Convert the polar point $(8, 150^\circ)$ to rectangular coordinates.

$x = 8 \cos 150^\circ = 8 \left(-\frac{\sqrt{3}}{2}\right) = -4\sqrt{3}$, $y = 8 \sin 150^\circ = 8 \left(\frac{1}{2}\right) = 4$. So the point is $(-4\sqrt{3}, 4)$.

- 8 Convert the polar point $\left(6, \frac{\pi}{3}\right)$ to rectangular coordinates.

$x = 6 \cos \frac{\pi}{3} = 6 \left(\frac{1}{2}\right) = 3$, $y = 6 \sin \frac{\pi}{3} = 6 \left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$. So the point is $(3, 3\sqrt{3})$.

Rectangular to polar conversion

- 9 Convert the rectangular point $(-3, 3\sqrt{3})$ to polar coordinates (r, θ) with $r > 0$ and $0^\circ \leq \theta < 360^\circ$.

$$r = \sqrt{(-3)^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = 6.$$

Since the point is in Quadrant II, $\theta = 120^\circ$ (from $\tan \theta = \frac{3\sqrt{3}}{-3} = -\sqrt{3}$, adjusted for the correct quadrant).

Multiple choice — easy

- 10 If $\sin x = \frac{12}{37}$ and x is in Quadrant I (so $\cos x = \frac{35}{37}$), find $\sin 2x$.

- a. $\frac{840}{1369}$ ✓
- b. $\frac{1081}{1369}$
- c. $-\frac{1081}{1369}$
- d. $-\frac{840}{1369}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{12}{37}\right)\left(\frac{35}{37}\right) = \frac{840}{1369}.$$

- 11 If $\sin x = \frac{11}{61}$ and x is in Quadrant I (so $\cos x = \frac{60}{61}$), find $\sin 2x$.

- a. $-\frac{1320}{3721}$
- b. $\frac{3479}{3721}$
- c. $\frac{1320}{3721}$ ✓
- d. $-\frac{3479}{3721}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{11}{61}\right)\left(\frac{60}{61}\right) = \frac{1320}{3721}.$$

- 12 If $\sin x = \frac{7}{25}$ and x is in Quadrant I (so $\cos x = \frac{24}{25}$), find $\sin 2x$.

- a. $-\frac{527}{625}$
- b. $\frac{527}{625}$
- c. $-\frac{336}{625}$
- d. $\frac{336}{625}$ ✓

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{7}{25}\right)\left(\frac{24}{25}\right) = \frac{336}{625}.$$

- 13 If $\sin x = \frac{8}{17}$ and x is in Quadrant I (so $\cos x = \frac{15}{17}$), find $\sin 2x$.

- a. $\frac{161}{289}$
- b. $\frac{240}{289}$ ✓
- c. $-\frac{161}{289}$
- d. $-\frac{240}{289}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{8}{17}\right)\left(\frac{15}{17}\right) = \frac{240}{289}.$$

- 14 If $\sin x = \frac{15}{17}$ and x is in Quadrant I (so $\cos x = \frac{8}{17}$), find $\sin 2x$.

- a. $\frac{161}{289}$
- b. $-\frac{161}{289}$
- c. $-\frac{240}{289}$
- d. $\frac{240}{289}$ ✓

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{15}{17}\right)\left(\frac{8}{17}\right) = \frac{240}{289}.$$

- 15 If $\sin x = \frac{3}{5}$ and x is in Quadrant I (so $\cos x = \frac{4}{5}$), find $\sin 2x$.

- a. $-\frac{7}{25}$
- b. $-\frac{24}{25}$
- c. $\frac{7}{25}$
- d. $\frac{24}{25}$ ✓

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}.$$

- 16 If $\sin x = \frac{5}{13}$ and x is in Quadrant I (so $\cos x = \frac{12}{13}$), find $\sin 2x$.

- a. $-\frac{120}{169}$
- b. $-\frac{119}{169}$
- c. $\frac{119}{169}$
- d. $\frac{120}{169}$ ✓

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{5}{13}\right)\left(\frac{12}{13}\right) = \frac{120}{169}.$$

- 17 If $\sin x = \frac{4}{5}$ and x is in Quadrant I (so $\cos x = \frac{3}{5}$), find $\sin 2x$.

a. $\frac{24}{25}$ ✓

b. $\frac{7}{25}$

c. $-\frac{7}{25}$

d. $-\frac{24}{25}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{4}{5}\right)\left(\frac{3}{5}\right) = \frac{24}{25}.$$

- 18 If $\sin x = \frac{9}{41}$ and x is in Quadrant I (so $\cos x = \frac{40}{41}$), find $\sin 2x$.

a. $\frac{720}{1681}$ ✓

b. $-\frac{1519}{1681}$

c. $-\frac{720}{1681}$

d. $\frac{1519}{1681}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{9}{41}\right)\left(\frac{40}{41}\right) = \frac{720}{1681}.$$

- 19 If $\sin x = \frac{60}{61}$ and x is in Quadrant I (so $\cos x = \frac{11}{61}$), find $\sin 2x$.

a. $\frac{3479}{3721}$

b. $-\frac{3479}{3721}$

c. $\frac{1320}{3721}$ ✓

d. $-\frac{1320}{3721}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{60}{61}\right)\left(\frac{11}{61}\right) = \frac{1320}{3721}.$$

- 20 If $\sin x = \frac{20}{29}$ and x is in Quadrant I (so $\cos x = \frac{21}{29}$), find $\sin 2x$.

- a. $\frac{41}{841}$
- b. $\frac{840}{841}$ ✓
- c. $-\frac{840}{841}$
- d. $-\frac{41}{841}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{20}{29}\right)\left(\frac{21}{29}\right) = \frac{840}{841}.$$

- 21 If $\sin x = \frac{21}{29}$ and x is in Quadrant I (so $\cos x = \frac{20}{29}$), find $\sin 2x$.

- a. $-\frac{840}{841}$
- b. $\frac{840}{841}$ ✓
- c. $\frac{41}{841}$
- d. $-\frac{41}{841}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{21}{29}\right)\left(\frac{20}{29}\right) = \frac{840}{841}.$$

- 22 If $\sin x = \frac{40}{41}$ and x is in Quadrant I (so $\cos x = \frac{9}{41}$), find $\sin 2x$.

- a. $-\frac{1519}{1681}$
- b. $\frac{1519}{1681}$
- c. $-\frac{720}{1681}$
- d. $\frac{720}{1681}$ ✓

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{40}{41}\right)\left(\frac{9}{41}\right) = \frac{720}{1681}.$$

- 23 If $\sin x = \frac{24}{25}$ and x is in Quadrant I (so $\cos x = \frac{7}{25}$), find $\sin 2x$.

- a. $-\frac{336}{625}$
- b. $\frac{527}{625}$
- c. $-\frac{527}{625}$
- d. $\frac{336}{625}$ ✓

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{24}{25}\right)\left(\frac{7}{25}\right) = \frac{336}{625}.$$

24 If $\sin x = \frac{12}{13}$ and x is in Quadrant I (so $\cos x = \frac{5}{13}$), find $\sin 2x$.

a. $-\frac{120}{169}$

b. $\frac{120}{169}$ ✓

c. $-\frac{119}{169}$

d. $\frac{119}{169}$

$$\sin 2x = 2\sin x \cos x = 2\left(\frac{12}{13}\right)\left(\frac{5}{13}\right) = \frac{120}{169}.$$

Multiple choice — medium

25 A triangle has $a = 14$, $b = 14$, and included angle $C = 60^\circ$. Find c .

a. 15

b. 13

c. 14 ✓

d. 20

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 196 + 196 - 196 = 196, \text{ so } c = \sqrt{196} = 14.$$

26 A triangle has $a = 12$, $b = 12$, and included angle $C = 60^\circ$. Find c .

a. 17

b. 13

c. 11

d. 12 ✓

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 144 + 144 - 144 = 144, \text{ so } c = \sqrt{144} = 12.$$

27 A triangle has $a = 14$, $b = 16$, and included angle $C = 120^\circ$. Find c .

- a. 21
- b. 25
- c. 26 ✓
- d. 27

$$c^2 = a^2 + b^2 - 2ab\cos(120^\circ) = 196 + 256 + 224 = 676, \text{ so } c = \sqrt{676} = 26.$$

28 A triangle has $a = 3$, $b = 5$, and included angle $C = 120^\circ$. Find c .

- a. 9
- b. 8
- c. 6
- d. 7 ✓

$$c^2 = a^2 + b^2 - 2ab\cos(120^\circ) = 9 + 25 + 15 = 49, \text{ so } c = \sqrt{49} = 7.$$

29 A triangle has $a = 17$, $b = 17$, and included angle $C = 60^\circ$. Find c .

- a. 16
- b. 18
- c. 24
- d. 17 ✓

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 289 + 289 - 289 = 289, \text{ so } c = \sqrt{289} = 17.$$

30 A triangle has $a = 13$, $b = 13$, and included angle $C = 60^\circ$. Find c .

- a. 12
- b. 14
- c. 18
- d. 13 ✓

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 169 + 169 - 169 = 169, \text{ so } c = \sqrt{169} = 13.$$

31 A triangle has $a = 5$, $b = 12$, and included angle $C = 90^\circ$. Find c .

a. 13 ✓

b. 12

c. 15

d. 14

$$c^2 = a^2 + b^2 - 2ab\cos(90^\circ) = 5^2 + 12^2 = 25 + 144 = 169, \text{ so } c = \sqrt{169} = 13.$$

32 A triangle has $a = 9$, $b = 9$, and included angle $C = 60^\circ$. Find c .

a. 13

b. 8

c. 10

d. 9 ✓

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 81 + 81 - 81 = 81, \text{ so } c = \sqrt{81} = 9.$$

33 A triangle has $a = 16$, $b = 16$, and included angle $C = 60^\circ$. Find c .

a. 16 ✓

b. 23

c. 15

d. 17

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 256 + 256 - 256 = 256, \text{ so } c = \sqrt{256} = 16.$$

34 A triangle has $a = 6$, $b = 8$, and included angle $C = 90^\circ$. Find c .

a. 11

b. 10 ✓

c. 9

d. 12

$$c^2 = a^2 + b^2 - 2ab\cos(90^\circ) = 6^2 + 8^2 = 36 + 64 = 100, \text{ so } c = \sqrt{100} = 10.$$

35 A triangle has $a = 6$, $b = 6$, and included angle $C = 60^\circ$. Find c .

a. 5

b. 6 ✓

c. 8

d. 7

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 36 + 36 - 36 = 36, \text{ so } c = \sqrt{36} = 6.$$

36 A triangle has $a = 9$, $b = 15$, and included angle $C = 120^\circ$. Find c .

a. 17

b. 21 ✓

c. 20

d. 22

$$c^2 = a^2 + b^2 - 2ab\cos(120^\circ) = 81 + 225 + 135 = 441, \text{ so } c = \sqrt{441} = 21.$$

37 A triangle has $a = 5$, $b = 5$, and included angle $C = 60^\circ$. Find c .

a. 4

b. 6

c. 7

d. 5 ✓

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 25 + 25 - 25 = 25, \text{ so } c = \sqrt{25} = 5.$$

38 A triangle has $a = 10$, $b = 10$, and included angle $C = 60^\circ$. Find c .

a. 14

b. 10 ✓

c. 11

d. 9

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 100 + 100 - 100 = 100, \text{ so } c = \sqrt{100} = 10.$$

39 A triangle has $a = 5$, $b = 8$, and included angle $C = 60^\circ$. Find c .

a. 8

b. 6

c. 9

d. 7 ✓

$$c^2 = a^2 + b^2 - 2ab\cos(60^\circ) = 25 + 64 - 40 = 49, \text{ so } c = \sqrt{49} = 7.$$

Multiple choice — hard

40 In a triangle, $\angle A = 135^\circ$, $a = 16$, and $\angle B = 45^\circ$. Find b .

a. 17

b. 32

c. 16 ✓

d. 18

$$\text{By the Law of Sines, } \frac{b}{\sin B} = \frac{a}{\sin A}, \text{ so } b = a \cdot \frac{\sin 45^\circ}{\sin 135^\circ} = 16(1) = 16.$$

41 In a triangle, $\angle A = 90^\circ$, $a = 8$, and $\angle B = 30^\circ$. Find b .

a. 4 ✓

b. 6

c. 8

d. 5

By the Law of Sines, $\frac{b}{\sin B} = \frac{a}{\sin A}$, so $b = a \cdot \frac{\sin 30^\circ}{\sin 90^\circ} = 8(1/2) = 4$.

42 In a triangle, $\angle A = 135^\circ$, $a = 3$, and $\angle B = 45^\circ$. Find b .

a. 3 ✓

b. 6

c. 4

d. 5

By the Law of Sines, $\frac{b}{\sin B} = \frac{a}{\sin A}$, so $b = a \cdot \frac{\sin 45^\circ}{\sin 135^\circ} = 3(1) = 3$.

43 In a triangle, $\angle A = 135^\circ$, $a = 17$, and $\angle B = 45^\circ$. Find b .

a. 19

b. 18

c. 34

d. 17 ✓

By the Law of Sines, $\frac{b}{\sin B} = \frac{a}{\sin A}$, so $b = a \cdot \frac{\sin 45^\circ}{\sin 135^\circ} = 17(1) = 17$.

44 In a triangle, $\angle A = 30^\circ$, $a = 12$, and $\angle B = 90^\circ$. Find b .

a. 12

b. 48

c. 24 ✓

d. 25

By the Law of Sines, $\frac{b}{\sin B} = \frac{a}{\sin A}$, so $b = a \cdot \frac{\sin 90^\circ}{\sin 30^\circ} = 12(2) = 24$.

45 Convert the rectangular point $(7, 24)$ to polar coordinates (r, θ) with $r > 0$. Find r .

a. 24

b. 27

c. 7

d. 25 ✓

$$r = \sqrt{(7)^2 + (24)^2} = \sqrt{49 + 576} = \sqrt{625} = 25.$$

46 Convert the rectangular point $(3, -4)$ to polar coordinates (r, θ) with $r > 0$. Find r .

a. 7

b. 3

c. 5 ✓

d. 4

$$r = \sqrt{(3)^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

47 Convert the rectangular point $(12, 35)$ to polar coordinates (r, θ) with $r > 0$. Find r .

a. 37 ✓

b. 39

c. 35

d. 12

$$r = \sqrt{(12)^2 + (35)^2} = \sqrt{144 + 1225} = \sqrt{1369} = 37.$$

48 Convert the rectangular point $(-8, -15)$ to polar coordinates (r, θ) with $r > 0$. Find r .

a. 17 ✓

b. 15

c. 8

d. 19

$$r = \sqrt{(-8)^2 + (-15)^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$$

49 Convert the rectangular point $(12, -35)$ to polar coordinates (r, θ) with $r > 0$. Find r .

a. 12

b. 37 ✓

c. 35

d. 39

$$r = \sqrt{(12)^2 + (-35)^2} = \sqrt{144 + 1225} = \sqrt{1369} = 37.$$

50 If $\sin x = \frac{12}{37}$, find $\cos 2x$.

a. $\frac{840}{1369}$

b. $-\frac{840}{1369}$

c. $-\frac{1081}{1369}$

d. $\frac{1081}{1369}$ ✓

$$\cos 2x = 1 - 2\sin^2 x = 1 - 2\left(\frac{12}{37}\right)^2 = 1 - \frac{288}{1369} = \frac{1081}{1369}.$$

51 If $\sin x = \frac{8}{17}$, find $\cos 2x$.

a. $\frac{161}{289}$ ✓

b. $\frac{240}{289}$

c. $-\frac{240}{289}$

d. $-\frac{161}{289}$

$$\cos 2x = 1 - 2\sin^2 x = 1 - 2\left(\frac{8}{17}\right)^2 = 1 - \frac{128}{289} = \frac{161}{289}.$$

52 If $\sin x = \frac{7}{25}$, find $\cos 2x$.

a. $-\frac{527}{625}$

b. $-\frac{336}{625}$

c. $\frac{527}{625}$ ✓

d. $\frac{336}{625}$

$$\cos 2x = 1 - 2\sin^2 x = 1 - 2\left(\frac{7}{25}\right)^2 = 1 - \frac{98}{625} = \frac{527}{625}.$$

53 If $\sin x = \frac{35}{37}$, find $\cos 2x$.

a. $\frac{840}{1369}$

b. $\frac{1081}{1369}$

c. $-\frac{1081}{1369}$ ✓

d. $-\frac{840}{1369}$

$$\cos 2x = 1 - 2\sin^2 x = 1 - 2\left(\frac{35}{37}\right)^2 = 1 - \frac{2450}{1369} = -\frac{1081}{1369}.$$

54 If $\sin x = \frac{15}{17}$, find $\cos 2x$.

a. $-\frac{240}{289}$

b. $\frac{161}{289}$

c. $-\frac{161}{289}$ ✓

d. $\frac{240}{289}$

$$\cos 2x = 1 - 2\sin^2 x = 1 - 2\left(\frac{15}{17}\right)^2 = 1 - \frac{450}{289} = -\frac{161}{289}.$$